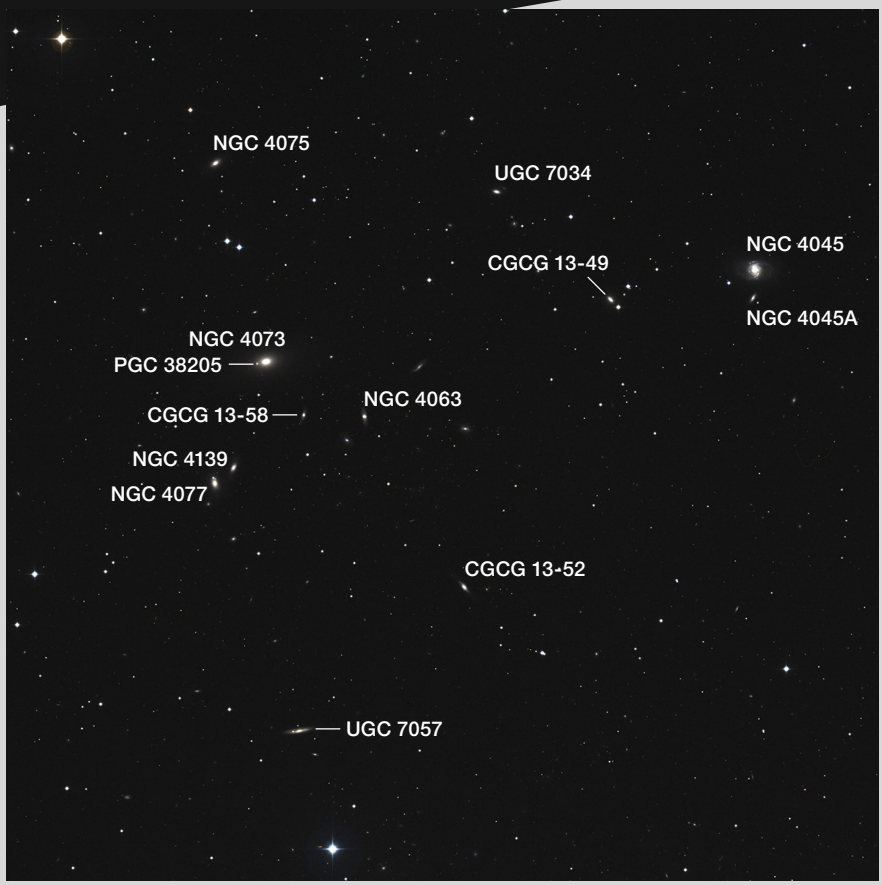

◀ **TODD'S TARGETS** In his search for a planet beyond Neptune, American astronomer David Todd discovered numerous faint galaxies. He also missed some. Use the image at left to locate the galaxies described in the text.

David Todd's Accidental Deep-Sky Discoveries

While searching for a distant planet, this American astronomer stumbled upon several galaxies in Leo and Virgo.

On February 6, 1878, David Peck Todd discovered the 14th-magnitude galaxy IC 591 using the U.S. Naval Observatory's 26-inch Clark refractor. The galaxy's location is surprising, just 23′ northwest of 1.4-magnitude Regulus, or Alpha (α) Leonis. But Todd wasn't seeking new nebulae — his aim was to discover a planet beyond Neptune! He conducted a little-known visual search and, as we'll see, in the process dug up some intriguing galaxies.

Todd wasn't the first to predict a new, distant planet. In 1848, just two years after the discovery of Neptune, French scientist Jacques Babinet proposed a trans-Neptunian planet. He arrived at this conclusion by analyzing perturbations in Neptune's motion using the predicted orbital elements by French mathematician Urbain Le Verrier and English astronomer John Couch Adams. Babinet dubbed the unseen world Hyperion.

Two years later, James Ferguson, an astronomer at the Naval Observatory, measured the asteroid Hygiea's position relative to a nearby 9th-magnitude star. English astronomer John Russell Hind created a stir when he looked for the reference star and came up empty. Ferguson took up the search but also failed to retrieve it, leading to speculation it might instead be an unknown planet. In 1878, asteroid hunter Christian Heinrich Friedrich Peters solved the mystery of the missing star. A mislabeled micrometer wire caused Ferguson to measure an errant position. By this date, however, the prospect of a trans-Neptunian world had captured other astronomers' imaginations.

Todd reanalyzed the residuals for the positions of Uranus and Neptune in 1877. His calculations pointed to a trans-Neptunian planet of magnitude 13 or fainter at a distance of 52 astronomical units (a.u.). Based on an expected longitude of 170°±10°, Todd began a visual search in Leo on November 3, 1877. He carefully examined each field at 400× to 600×, covering 40° along the ecliptic, and checked any nonstellar objects for movement. The project ended unsuccessfully on March 6, 1878.

Did he stand a chance in his pursuit? As far as stumbling upon Pluto, it was in Taurus at the time, a full 14° south of the ecliptic and far beyond his search region. Furthermore, he was looking for a tiny but distinct disk, estimated at 2.1″ in diameter. Pluto, of course, appears stellar in a telescope.

So, what did he find? A modern analysis shows Todd netted 28 galaxies, including several previously known. Many others, though, were unidentified, as his rough positions using the refractor's setting circles landed on empty patches of sky. But by reviewing his sketches and positional offsets from nearby stars, catalog sleuths Harold Corwin, Wolfgang Steinicke, and others (including myself) recovered 16 of the 28 as original discoveries.

"Suspected Object"

Let's examine the evidence for **IC 591**, which Todd described as a "Suspected object faint and diffused." A finder

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diagram includes Regulus and Uranus, which happened to be very nearby. His eyepiece sketch from February 6th (below) shows an oval object labeled xxii b, along with a star (xxii a) preceding (west) by 86 seconds of time, or 21′.

How can we confirm Todd's observation of IC 591 without a precise position? Although he only included five stars in his diagram, they match up perfectly with the sky, and a 13.4-magnitude star 21′ to the west of the galaxy's core is the clincher. He missed Leo I, which is only 15′ east of IC 591. With a magnification of 400×, his scope's narrow field was too small to frame this large, low-surface-brightness dwarf galaxy (see *S&T:* Oct. 2023, p. 57). John Louis Emil Dreyer skipped Todd's nebula when he compiled the *New General Catalogue of Nebulae and Clusters of Stars*, published in 1888. However, French astronomer Stéphane Javelle rediscovered it in 1892, and he received discovery credit in the *Index Catalog*.

"Large and Nebulous"

In late December 1877 and early January 1878, Todd perused star fields in western Virgo, about 3½° east of Beta (β) Virginis, and ran across the galaxy group now called MKW 4. A redshift survey published in 2002 lists 50 members within a circle 2.4° in diameter. William Herschel first swept this region in 1784 with his 18.7-inch speculum-metal reflector and discovered **NGC 4073**, at magnitude 11.5 the brightest member of the group, along with 12th-magnitude **NGC 4045** and 13th-magnitude **NGC 4077**.

Todd first picked up 13.5-magnitude **NGC 4075** (object xi) on December 27, 1877, noting "a small, but poorly defined disk." My 18-inch Dob shows it as a modest oval tipped toward the southwest with a starlike nucleus. The following week, he encountered NGC 4073 (xii a) and called it "large and nebulous." Through my 18-inch, this 11.5-magnitude giant elliptical is well concentrated and increases to a small, prominent nucleus. Nearby, he uncovered **NGC 4063** (xii d), a 14th-magnitude galaxy with a weak concentration. Once again, Todd lost out on discovery recognition, this time to French astronomer Édouard Stephan (of Stephan's Quintet fame), who spotted it four months later but measured a precise position.

On January 2, 1878, Todd shifted his gaze less than ½° west-northwest and came upon more nonstellar objects. NGC 4045 (xiii b) is a foreground, 12th-magnitude spiral with active star formation in the nucleus and a circular inner ring. I noticed a roundish core and distinct stellar nucleus within a slightly elongated envelope. Todd's drawing shows a faint star (xiii a) about 1½′ south of NGC 4045. Apparently, he mistook the 14th-magnitude galaxy **NGC 4045A** (CGCG 13-45) for a star, though I saw it as small, pale, and somewhat elongated. Todd discovered another faint nebula, the lenticular galaxy **CGCG 13-49** (xiii e), 8′ east of NGC 4045 and just northeast of an 11th-magnitude star (xiii d). My 18-inch reveals a fuzzy 25″ oval extending in the star's direction with a string of three 11th- to 13th-magnitude stars trailing to the northwest.

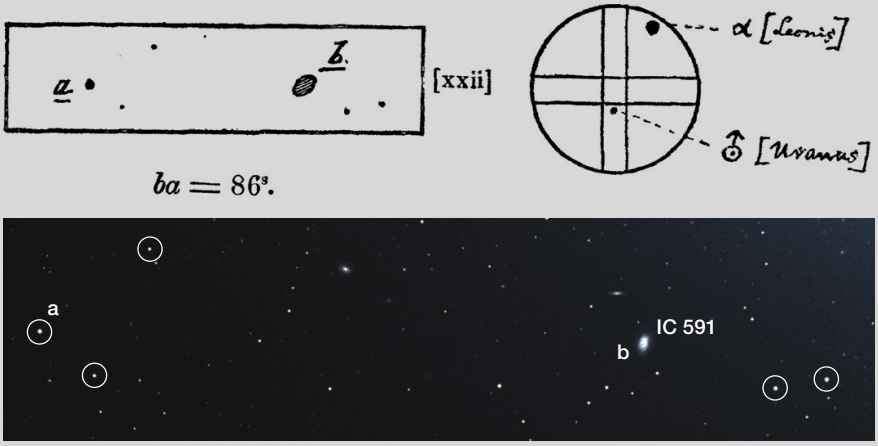
Three nights later, Todd returned to this field and recorded NGC 4077 (xiv a) as a potential candidate. When the seeing improved the next night, he noted "a star with faint nebulous border." The "star" is likely the galaxy's nucleus. I logged NGC 4077 as elongated north-to-south with a 14.7-magnitude star pinned to the north edge of its halo. In addition, he found **NGC 4139** (xiv b), a smaller and fainter lenticular (magnitude 13.8) only 1.3′ to the northwest. In my 18-inch, the oval halo spans 30″ in diameter and intensifies at the center. Astronomers sometimes identify this galaxy as IC 2989 — a designation resulting from an erroneous NGC position.

LISTENING TO MARS

Todd was a vocal advocate for the notion of intelligent life on Mars. On August 21–23, 1924, with Mars in opposition, he promoted a "National Radio Silence Day," which urged citizens to maintain radio silence for the first five minutes of every hour during a 36-hour period. Army radio stations listened for signals, and a cryptographer was even on hand to translate any messages from the Red Planet. Needless to say, none arrived.

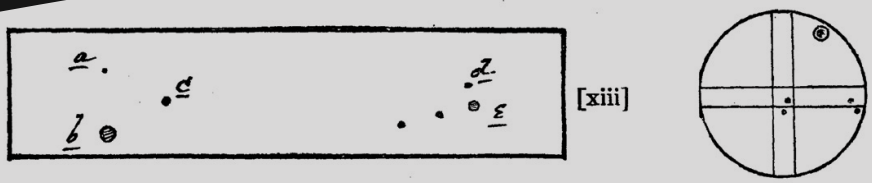
Todd's Misses

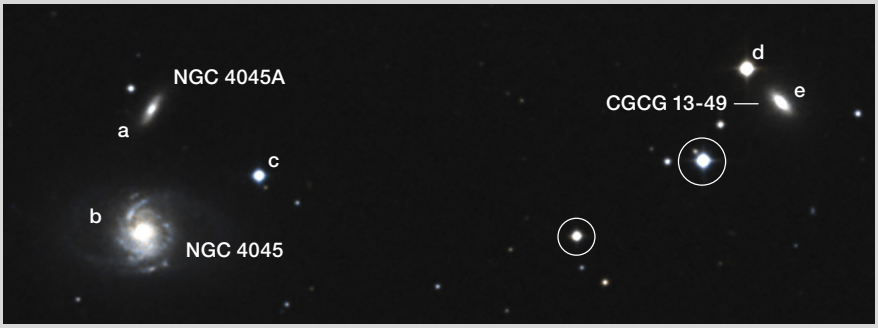
Last April, I revisited this cluster with my 24-inch f/3.8 reflector at 375× and viewed several galaxies that Todd missed in his survey. The brightest is **UGC 7034**, situated 14′ east-



▲ **STUMBLING UPON IC 591** This sketch and the one on page 60 are taken from an article Todd wrote describing his search for the elusive trans-Neptunian planet (published in the *Proceedings of the American Academy of Arts and Sciences* in 1885, vol. 21). His diagrams are all south up, and the accompanying images have the same orientation.

TODD, DAVID P. / *PROCEEDINGS OF THE AMERICAN ACADEMY OF ARTS AND SCIENCES* / 1885; SDSS; POSS-II / STSCI / CALTECH / PALOMAR OBSERVATORY





▶ **NGC 4045 AND FAINT COMPANIONS**
Todd took meticulous notes during his months-long search for the distant planet. He drew detailed diagrams that he'd return to later when he'd reobserve the field, to check if any target had moved in the intervening time.

northeast of NGC 4045. This small, 14th-magnitude lenticular bulges slightly from west to east. **CGCG 13-58** is a 14.6-magnitude patch only 3.5′ southwest of NGC 4073. Shifting 13′ farther southwest, I located 14th-magnitude **CGCG 13-52**, a thin, low-surface-brightness slash with its central axis pointing back towards NGC 4073. The narrow edge-on galaxy **UGC 7057** is on the southern side of the cluster, less than 7′ north-northeast of a 7.7-magnitude star. With the star placed outside the field, the 13.6-magnitude galaxy was a ghostly, uniform streak.

One member of the cluster is quite easy to overlook. Virtually stellar, **PGC 38205** is camouflaged in the halo of NGC 4073, just 30″ east-southeast of center. This 15th-magnitude compact elliptical may be the remnant core of a much larger galaxy that was stripped of its outer stars by NGC 4073. M32 (M31's much smaller companion) is considered the prototype of this rare class.

Two years after Todd finished his search, George Forbes, an astronomy professor at Glasgow University in Scotland, proposed two new outer planets at 100 and 300 a.u. Forbes argued these planets diverted several known comets into elliptical orbits with similar aphelion distances. Todd was excited to read that Forbes's predicted position for the interior planet differed by only 4° from his position. He wrote a general account of his investigation in 1880 and a detailed report in 1885, expressing hope it would inspire a thorough, deeper survey with a telescope equipped for dry-plate photography. Pioneering Welsh astrophotographer Isaac Roberts took up the cause in 1890 from his new observatory in Crowborough, Sussex (England). He compared two sets of 18 photographic plates taken at least a week apart in the same region, but also struck out.

In 1881, Todd was appointed Director of Massachusetts's Amherst College Observatory, where he remained a professor of astronomy until retiring in 1917. While at Amherst, he became friends with Percival Lowell, who financed an expedition led by Todd to observe the May 28, 1900, total solar eclipse from Tripoli, Libya. Lowell recruited Todd again in 1907 to head an expedition to Chile's high desert, hoping to obtain photographic proof of Martian canals. In return, Todd's visual pursuit of a ninth planet perhaps motivated Lowell, who recalculated a theoretical position and began his first photographic search for a trans-Neptunian world in 1905.

■ The history of deep-sky discovery continues to fascinate Contributing Editor **STEVE GOTTLIEB**, who has been writing for *Sky & Telescope* for the past 25 years. You can reach him at **astrogottlieb@gmail.com**.

Todd's Targets and More

Object	Type	Todd desig.	Mag(v)	Size	RA	Dec.
IC 591	Barred sp.	xxii	14.0	1.0′ × 0.7′	10^h 07.5^m	+12° 16′
NGC 4073	Elliptical	xii a	11.5	3.2′ × 2.3′	12^h 04.4^m	+01° 54′
NGC 4045	Spiral	xiii b	12.0	2.7′ × 1.9′	12^h 02.7^m	+01° 59′
NGC 4077	Lenticular	xiv a	13.2	1.3′ × 0.9′	12^h 04.6^m	+01° 47′
NGC 4075	Lenticular	xi	13.5	1.2′ × 0.6′	12^h 04.6^m	+02° 04′
NGC 4063	Lenticular	xii d	14.0	1.1′ × 0.5′	12^h 04.1^m	+01° 51′
NGC 4045A	Lenticular	xiii a	14.3	0.6′ × 0.4′	12^h 02.7^m	+01° 57′
CGCG 13-49	Lenticular	xiii e	13.7	0.8′ × 0.6′	12^h 03.2^m	+01° 57′
NGC 4139	Lenticular	xiv b	13.8	1.0′ × 0.5′	12^h 04.6^m	+01° 48′
UGC 7034	Lenticular	—	13.9	1.0′ × 0.4′	12^h 03.6^m	+02° 03′
CGCG 13-58	Lenticular	—	14.6	0.8′ × 0.3′	12^h 04.3^m	+01° 51′
CGCG 13-52	Spiral	—	13.8	1.0′ × 0.4′	12^h 03.7^m	+01° 42′
UGC 7057	Spiral	—	13.6	1.9′ × 0.4′	12^h 04.3^m	+01° 34′
PGC 38205	Elliptical	—	~15	—	12^h 04.5^m	+01° 54′

Angular sizes are from recent catalogs. Visually, an object's size is often smaller than the cataloged value and varies according to the aperture and magnification of the viewing instrument. Right ascension and declination are for equinox 2000.0.

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